

Exercise

CONSIDER THE FOLLOWING DATASET:

	0	1	2	3
$u(t)$	0	1	1	1
$y(t)$	1	1	2	$3/2$

AND THIS MODEL CLASS

$$M(\theta): y(t) = a y(t-1) + u(t-1) + \delta(t) \quad \delta(t) \sim \text{wn}(\mu, \lambda^2)$$

⇒ IDENTIFY THE MODEL PARAMETERS: a, μ, λ^2

STEPS:

1. COMPUTE THE PREDICTOR
2. COMPUTE SAMPLES OF $\hat{y}(t+1|t)$ $t=1, \dots, 4$ AS FUNCTION OF a, b, μ
3. COMPUTE $J_N(a, b, \mu)$
4. MINIMIZE $J_N(a, b, \mu)$

1. PREDICTOR

$$y(t) = \frac{\boxed{1}^{B_1}}{1 - az^{-1}} u(t-1) + \frac{\boxed{1}^{C(z)}}{1 - az^{-1}} y(t) + \frac{\boxed{1}^{B_2(z)}}{1 - az^{-1}} \mu$$

$$\Downarrow$$

$$\hat{y}(t+1|t) = \frac{1}{1} u(t-1+1) + \frac{az^{-1}}{1} y(t+1) + \frac{1}{1} \mu$$

$$\boxed{\hat{y}(t+1|t) = u(t) + a y(t) + \mu}$$

2. SAMPLES OF $\hat{y}(t+1|t)$

$$\begin{aligned} \hat{y}(1) &= 0 + a \cdot y(0) + \mu = a + \mu \\ \hat{y}(2) &= 1 + a \cdot y(1) + \mu = 1 + a + \mu \\ \hat{y}(3) &= 1 + a \cdot y(2) + \mu = 1 + 2a + \mu \end{aligned}$$

3. COST FUNCTION

$$\begin{aligned} J_N &= \frac{1}{N} \sum_{t=1}^3 (\hat{y}(t) - y(t))^2 \\ &= \frac{1}{3} \left[(a + \mu - 1)^2 + (1 + a + \mu - 2)^2 + (1 + 2a + \mu - 3/2)^2 \right] \\ &= \frac{1}{3} \left[(a + \mu - 1)^2 + (a + \mu - 1) + (2a + \mu - 1/2)^2 \right] \\ &= \frac{1}{3} \left[2(a + \mu - 1)^2 + (2a + \mu - 1/2)^2 \right] \end{aligned}$$

4. COST FUNCTION MINIMIZATION

$$\begin{aligned}\frac{\partial J_n}{\partial a} &= \frac{1}{3} [4(a+\mu-1) + 4(2a+\mu-\frac{1}{2})] \\ &= \frac{1}{3} [12a + 8\mu - 6]\end{aligned}$$

$$\begin{aligned}\frac{\partial J_n}{\partial \mu} &= \frac{1}{3} [4(a+\mu-1) + 2(2a+\mu-\frac{1}{2})] \\ &= \frac{1}{3} [8a + 6\mu - 5]\end{aligned}$$

WE HAVE A LINEAR SYSTEM OF EQUATIONS TO SOLVE!

$$\begin{aligned}12a + 8\mu &= 6 \\ 8a + 6\mu &= 5\end{aligned} \Rightarrow \begin{bmatrix} 12 & 8 \\ 8 & 6 \end{bmatrix} \Theta = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

$$\begin{aligned}\Theta &= \frac{1}{72 - 64} \cdot \begin{bmatrix} 6 & -8 \\ -8 & 12 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \end{bmatrix} \\ &= \frac{1}{8} \begin{bmatrix} 36 - 40 \\ -48 + 60 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 3/2 \end{bmatrix}\end{aligned}$$